Akoizo Original Practice - Form 3 - Answer key

Division B | Ping Pong Parachute | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) How may fins be attached to the bottle?

Unit: Building a legal, stable rocket.
Worked-case data: A rocket's CG is 18 cm from the nose and its CP is 21 cm from the nose. The bottle diameter is 8 cm. Is it stable enough?
Separate investigation observations:
Flight 1: Rocket corkscrews and lands near a wall.
Inspection: One fin is twisted about 8° from the body line.
Flight 2: Fin realigned: rocket flies straight up.

A. With tape only

B. With hot glue

C. With epoxy

D. With metal screws

Correct: A

2. (2 points) A payload never separates; the rocket takes 9.00 s to land. The log is complete. What is the launch score?

A. 8.10

B. 9.00

C. 2.25

D. 7.65

Correct: A

3. (2 points) When may the parachute payload be placed on the rocket?

Unit: Deployment and separation.
Worked-case data: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling. What should change?
Separate investigation observations:
Cup depth 2 cm: Payload left during climb in 4 of 5 flights.
Cup depth 4 cm: Payload stayed attached in 3 of 5 flights.
Cup depth 3 cm: Payload released near apogee in 5 of 5 flights.

A. Before loading the rocket on the launcher

B. At any time

C. After the rocket is on the launcher, before pressurizing

D. After pressurizing

Correct: C

4. (2 points) Repeated 65 psi flights reach 22-25 ft. The ceiling is 24 ft. What should you do?

A. Raise the pressure to finish faster

B. Remove the fins

C. Choose a lower pressure so the highest flights clear 24 ft

D. Launch at 65 psi; the average is below 24 ft

Correct: C

5. (4 points) Explain how compressed air pushes the rocket upward.

Unit: How an air-powered bottle rocket climbs.
Worked-case data: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?
Separate investigation observations:
40 psi: Peak about 12 ft; time aloft 8.1 s.
55 psi: Peak about 18 ft; time aloft 12.4 s.
70 psi: Peak about 24 ft; time aloft 16.0 s in a 30 ft hall.

answer: The bottle’s pressurized air rushes out the nozzle downward; by Newton’s third law, the expelled air pushes back up on the rocket. (4 points).

6. (4 points) Why should outliers stay in the log?

Unit: The practice log and launch decisions.
Worked-case data: A log has 12 flights recording pressure, height, time, and fin count. Is it complete?
Separate investigation observations:
Log entries: 14 flights recorded.
Columns: Pressure, peak height, time aloft, parachute diameter.
Flights 1-3: Peak height column left blank.

answer: They show real reliability problems, such as failed openings, and explain variation; deleting them hides risk. (4 points).

7. (4 points) If the canopy diameter doubles, roughly how does descent time change, and why?

answer: Area quadruples, so terminal velocity halves and descent time roughly doubles from the same height. (4 points).

8. (4 points) Explain how fins keep a rocket pointing forward.

Unit: Building a legal, stable rocket.
Worked-case data: A rocket's CG is 18 cm from the nose and its CP is 21 cm from the nose. The bottle diameter is 8 cm. Is it stable enough?
Separate investigation observations:
Flight 1: Rocket corkscrews and lands near a wall.
Inspection: One fin is twisted about 8° from the body line.
Flight 2: Fin realigned: rocket flies straight up.

answer: When the rocket tilts, airflow pushes on the fins behind the CG, creating a turning force that swings the nose back into the wind. (4 points).

9. (4 points) Why are three timers used instead of one?

Unit: The event and its score: time aloft and multipliers.
Worked-case data: A team with a complete log records 14.20 s on launch 1 (clean) and 19.80 s on launch 2, but launch 2 grazed a light on the way up. What is its score?
Separate investigation observations:
Ceiling: 24 ft gym.
Log data: 60 psi: about 20 ft peak, 15 s aloft; 75 psi: about 26 ft peak.
Team plan: Launch at 75 psi for "more height."

answer: Human reaction times vary; taking the middle of three reduces the effect of one slow or fast timer. (4 points).

10. (4 points) Why rehearse the loading sequence?

Unit: Deployment and separation.
Worked-case data: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling.
What should change?
Separate investigation observations:
Cup depth 2 cm: Payload left during climb in 4 of 5 flights.
Cup depth 4 cm: Payload stayed attached in 3 of 5 flights.
Cup depth 3 cm: Payload released near apogee in 5 of 5 flights.

answer: The payload cannot be adjusted after loading, so consistent, quick packing prevents tangles and saves time in the 5-minute window. (4 points).

11. (6 points) Solve this case and explain your reasoning: Your chart shows 50 psi -> 17 ft, 60 psi -> 21 ft, 70 psi -> 24 ft, with about ±1.5 ft spread. The ceiling is 25 ft. Which pressure should you choose?

reasoning: 70 psi could reach 25.5 ft: too risky.
60 psi reaches 21 ± 1.5 ft, at most 22.5 ft: safe.
Interpolating, about 63 psi gives roughly 22 ft, still clearing 25 ft with margin. (3 points).

conclusion: Choose about 60-63 psi and justify it with the chart and the observed spread. (3 points).

12. (2 points) What pattern do these flights show?

Supplied synthetic observations:
40 psi: Peak about 12 ft; time aloft 8.1 s.
55 psi: Peak about 18 ft; time aloft 12.4 s.
70 psi: Peak about 24 ft; time aloft 16.0 s in a 30 ft hall.

A. Pressure has no effect on time aloft

B. Time aloft falls as pressure rises

C. Higher apogee gives more descent time, so time aloft rises with pressure

Correct: C

13. (2 points) How will this log be judged?

Supplied synthetic observations:
Log entries: 14 flights recorded.
Columns: Pressure, peak height, time aloft, parachute diameter.
Flights 1-3: Peak height column left blank.

A. Complete, because time aloft is recorded

B. Incomplete: too few parameters and missing heights

C. Complete, because there are 14 flights

Correct: B

14. (2 points) Define “Terminal velocity” in the context of Ping-Pong Parachute.

answer: Steady falling speed when drag equals weight. (2 points).

15. (2 points) Define “Center of pressure (CP)” in the context of Ping-Pong Parachute.

answer: Where aerodynamic forces effectively act. (2 points).

16. (2 points) Define “Apogee” in the context of Ping-Pong Parachute.

answer: The highest point of the flight. (2 points).

17. (2 points) Define “Incomplete log” in the context of Ping-Pong Parachute.

answer: Missing a parameter or under 10 flights; x 0.85. (2 points).

18. (2 points) Define “Air density (ρ)” in the context of Ping-Pong Parachute.

answer: About 1.2 kg/m^3 at room conditions. (2 points).